Tender: Development of Centralized Access Control Integration System — B. BIRDS (Border Incident Reporting & Documentation System) - Phase II
e-GP tender tracking no.: 1347267

ANNEX A TO COMPLIANCE STATEMENT

Compliance Notes — BIRDS (Border Incident Reporting & Documentation System) - Phase II

This annex explains how the offered system meets each of the 58 items of the Technical Specification, where the Technical Evaluation Committee can verify it in the application demo, and whether the item is already demonstrated or is delivered during implementation on BGB infrastructure.

1. Application demo

Demo address	https://birds-demo.onrender.com
API documentation	https://birds-demo.onrender.com/api/docs (OpenAPI 3.1)
Languages	English and Bangla (switch in the page header)
Data	Sample data only; no BGB operational data

Demo accounts (one per role):

Role	Username	Password	Unit / scope
HQ Administrator	hq_admin	Admin@123	HQ (all units)
Sector Officer	sector_off	Sector@123	Sector-7
Battalion Operator	batt_op	Batt@123	17 BGB
Verifier/Approver	verifier	Verify@123	Sector-7

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Role	Username	Password	Unit / scope
Auditor	auditor	Audit@123	HQ (read-only audit)

2. Status legend

Status	Meaning	Items
Demonstrated	Working in the application demo now.	32
Demonstrated; completed at deployment	Working in the demo; final configuration or a stated addition is completed when deployed on BGB infrastructure.	12
Delivered during implementation	Infrastructure, service or document delivered during implementation (for example backup, DR, load testing, manuals, training).	14

3. Compliance notes by item

Ser	Item	How the offered system complies	Where to verify	Status
1.	**System Type**	BIRDS is a complete web system for recording, verifying, tracking, analysing and reporting border incidents across HQ, Sector, Battalion and BOP levels, with workflow, audit trail, GIS, analytics, reports and administration.	The whole application at the demo address.	Demonstrated
2.	**Deployment Architecture**	Three-tier architecture: React web client (presentation layer) → Node.js/Express REST API (application layer) → PostgreSQL (database layer). Installed on BGB servers and reached through BGB's intranet over HTTPS; the API and database can be scaled independently.	Technical proposal (architecture); API description at /api/docs.	Demonstrated

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3.	**User Management**	Five default roles (HQ Administrator, Sector Officer, Battalion Operator, Verifier/Approver, Auditor) with a permission matrix editable in the Admin module. Every user has an access scope (HQ, a Sector or a Battalion); the server enforces permissions and scope on every request.	Admin › Users & roles and Permissions matrix (hq_admin). Sign in as batt_op to see only 17 BGB data.	Demonstrated
4.	**Authentication**	Username/password sign-in; passwords hashed with bcrypt; configurable password policy and account lockout; signed (RS256) session tokens. Directory sign-in is built as a pluggable provider and is enabled with BGB's Active Directory/LDAP details at deployment (see item 32).	Sign-in page; Admin › Security & system (password policy, directory sign-in panel).	Demonstrated; completed at deployment
5.	**Incident Registration**	Three-step electronic form (details, location, evidence) with draft saving, submission, editing and resubmission after a return, and a warning about possible duplicate reports.	New incident (batt_op); My drafts view on Incidents.	Demonstrated
6.	**Incident Classification**	Categories and sub-categories (incident types) are created, renamed and activated/deactivated in the Admin module; four severity levels; standard status set driven by the workflow.	Admin › Categories (hq_admin); the incident form drop-downs.	Demonstrated
7.	**Incident Details**	The form records date and time, location and border area, Sector/Battalion/BOP, category and type, severity, description, persons/entities involved, action taken and outcome.	Any incident page; New incident.	Demonstrated
8.	**Geographical Information**	Latitude/longitude from the device GPS or entered by hand; each incident has a map on its page; the GIS map shows all incidents the user may see.	Incident page map; GIS map.	Demonstrated
9.	**Border Sector Information**	Each incident is linked to its Sector, Battalion and BOP from a configurable organisational hierarchy (HQ → Sector → Battalion → BOP) that administrators extend in the Admin module. Further levels such as Region are set up to BGB's structure during implementation.	Admin › Org hierarchy; Sector/Battalion/BOP fields on the form.	Demonstrated; completed at deployment

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10.	**Evidence & Document Management**	Photographs, videos, scanned documents, PDFs, reports and other documents (Word, Excel, PowerPoint in current and older formats, OpenDocument, RTF) are attached to individual incidents: up to 50 MB per file, file type checked on the content, macro-enabled documents refused, encrypted at rest. Photographs, videos and PDFs are previewed in place; other documents are downloaded.	Evidence section of an incident; step 3 of New incident.	Demonstrated
11.	**Media Metadata**	Original file name, type, size, upload time, file last-modified time and SHA-256 fingerprint are kept for every file; for photographs the EXIF data (date/time taken, camera, GPS) and for office documents the document properties (title, author, last modified by, created, modified) are preserved and shown. An integrity check re-verifies the fingerprint on demand.	Evidence section of an incident (Verify integrity).	Demonstrated
12.	**Incident Timeline**	Every incident keeps an append-only history of creation, submission, review, verification, approval, returns and rejections with reasons, modifications (with the reason), evidence changes and closure. Investigation notes and updates are recorded as comments on the incident, dated and attributed, alongside the history. Database triggers prevent changes to history and comments.	Timeline and Comments on any incident page.	Demonstrated
13.	**Workflow Management**	Workflow Draft → Submitted → Reviewed → Verified → Approved → Closed, with Return and Reject (reason required); each action is limited to the authorised roles and scope, and nobody can act on an incident they filed. The overdue time per step is configurable. Workflow stages and authorities are configured to BGB's requirements during implementation.	Incident page action buttons (sector_off, verifier, hq_admin); Admin › Workflow.	Demonstrated; completed at deployment

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14.	**Verification & Approval**	Supervisory officers review, return, modify, verify, approve or reject. While a report is submitted, reviewed or verified, the officer who may act on it at that stage can modify it before approval: a reason is required, each changed field is recorded with its old and new value in the audit trail, the timeline shows the change and the filing officer is notified. Return sends the report back to the filing officer for correction with a recorded reason.	Incident page: "Modify report", Review, Return, Reject (sector_off); Verify, Approve (verifier).	Demonstrated
15.	**Incident Tracking**	Every incident gets a unique number BIRDS-<year>-NNNN generated by the server. Incidents are tracked by reference number, status, location, date, category and responsible unit.	Header search; Incidents page filters.	Demonstrated
16.	**Search Facility**	Filters for date range, Sector/Battalion/BOP, category, incident type (sub-category), severity, status and reference number or title, with views All / Needs my action / My incidents / My drafts, sorting and paging; filters are kept in the address so a search can be shared.	Incidents page › Filters.	Demonstrated
17.	**Dashboard**	Dashboard per role and scope: total, open, pending review, verified (passed verification), approved/closed and high/critical counts, overdue items, a work queue of items waiting for the user, category and severity charts and trends, with a period selector.	Dashboard (compare hq_admin with batt_op).	Demonstrated
18.	**GIS/Map Dashboard**	Map of incidents with one marker per BOP showing its number of incidents, coloured by the most severe; clicking a point zooms in on it with zoom and close buttons, and "Show all" opens its incidents in the register, returning to the same map view; clustering, heat map, filters by category, severity, status and date; framed on Bangladesh.	GIS map: click a point to zoom in; "Show all N incidents"; "Back to the map".	Demonstrated

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19.	**Analytics & Statistics**	Statistics by incident type, location (sector and BOP hotspots), time period (hour of day, weekday), unit, frequency and severity, including a severity-weighted risk score; every chart opens the matching incidents.	Analytics page: breakdowns, When incidents happen, Hotspots, Sector risk summary.	Demonstrated
20.	**Trend Analysis**	Trend charts by day, week, month, quarter and year, with comparison against the previous year.	Analytics › trend chart granularity.	Demonstrated
21.	**Reporting**	Four report templates (operational, statistical, high-threat, sector performance) with filters. Printed reports follow a formal layout (letterhead, classification marking, reference number, signature blocks, page numbers) and can include every incident in full with its evidence.	Reports page › Print / PDF.	Demonstrated
22.	**Export**	Reports and data export to PDF (print/save as PDF), Excel and CSV, limited to what the user is permitted to see; every export and print is recorded in the audit trail.	Reports, Analytics and Audit trail export buttons.	Demonstrated
23.	**Notifications**	In-app notifications, e-mail (SMTP) and phone/browser push for new submissions, items pending review/verification, returns, approvals, rejections and comments; overdue items are flagged. Escalation notifications and administrator-configurable notification rules are delivered during implementation.	Notification bell; Profile › notifications on this device.	Demonstrated; completed at deployment
24.	**Audit Trail**	Tamper-evident audit log (SHA-256 hash chain, verifiable on demand) of sign-in/out, failed sign-ins, creation, modification, workflow actions, deletions, evidence views/downloads, exports and prints; append-only at database level; searchable and exportable.	Audit trail (auditor or hq_admin) › Verify chain.	Demonstrated

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25.	**Data Integrity**	The filing officer edits a record only while Draft or Returned; before approval, a reviewer at the current stage may modify it with a recorded reason and a field-by-field audit entry (item 14); approved, closed and rejected records cannot be altered or deleted by anyone; invalid or concurrent transitions are refused; history and audit tables are protected by database triggers.	Try editing an approved incident; Audit trail shows DENIED entries.	Demonstrated
26.	**Data Security**	Role- and scope-based access control, secure headers, CORS restriction, rate limits, input validation on every request, short-lived tokens held in memory, and evidence encrypted at rest (AES-256-GCM).	Technical proposal (security); Admin › Security & system.	Demonstrated
27.	**Encryption**	All traffic over HTTPS/TLS (as on the demo); evidence, voice messages and photos are encrypted at rest. Database storage encryption is configured on BGB servers at deployment.	Demo address uses HTTPS.	Demonstrated; completed at deployment
28.	**Session Security**	Configurable idle timeout with automatic sign-out, single active session per user (a new sign-in ends the old one), server-side session checks on every request, and a list of own sessions with "sign out other sessions".	Admin › Security & system; Profile › sessions.	Demonstrated
29.	**Password Security**	Configurable minimum length, complexity (upper case, digit, special character), failed-login threshold and lockout duration; forced change after an administrator reset. Password expiry is added as a configurable setting during implementation.	Admin › Security & system › Password policy.	Demonstrated; completed at deployment
30.	**Database**	PostgreSQL relational database with transactions, foreign keys, check constraints, indexes and append-only triggers; backup and recovery under items 34–35.	Database documentation (to be delivered); technical proposal.	Demonstrated
31.	**API / Integration**	Secure REST API (/api/v1) with token authentication and the same permission checks as the user interface, documented in OpenAPI 3.1.	/api/docs on the demo.	Demonstrated

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32.	**AD/LDAP Integration**	Integration with BGB Active Directory/LDAP over LDAPS for central sign-in and user provisioning, configured with BGB's directory details during implementation; local accounts remain available as fallback if BGB wishes.	Admin › Security & system (directory sign-in panel, marked DEMO).	Delivered during implementation
33.	**File Storage**	Central encrypted file storage on local disk/NAS or S3-compatible object storage (e.g. MinIO), with per-incident storage limits and a configurable retention period; records past retention are flagged for review. The disposal rule is agreed with BGB.	Admin › Security & system › Archiving & retention.	Demonstrated; completed at deployment
34.	**Backup**	Automated, encrypted, scheduled backup of the database and evidence files to separate storage, with configurable frequency (setting already present in the Admin module), logged in the audit trail, with periodic restore tests.	Admin › Operations (marked DEMO).	Delivered during implementation
35.	**Disaster Recovery**	Deployment at a BGB-authorised DR site with database replication and evidence storage replication, and a documented, tested restoration procedure.	Backup/Recovery Procedure (to be delivered).	Delivered during implementation
36.	**High Availability**	The application holds no state in memory (sessions in the database, realtime events through the database), so several application servers can run behind a load balancer; the database runs with a standby replica where required.	Technical proposal (architecture).	Delivered during implementation
37.	**Scalability**	Stateless application servers, indexed database, object storage for files and paging on every list let users, incidents, files, units and history grow without redesign.	Technical proposal (architecture).	Demonstrated; completed at deployment
38.	**Concurrent Users**	The system is dimensioned for 500 concurrent users (the highest figure in the specification) with reserve capacity: one application process measured more than 650 requests per second, 95 % within 100 ms, against an estimated peak of about 350 for 500 users (Technical Proposal, section 7). A load test at 100, 250 and 500 users on the installed infrastructure is reported before acceptance.	Technical Proposal §7, §9; load-test report (to be delivered).	Delivered during implementation

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39.	**Performance**	Normal transactions respond within 3 seconds under the specified load; measured in the load test and reported.	Load-test report (to be delivered).	Delivered during implementation
40.	**Availability**	99.5 % availability through the high-availability setup (item 36), monitoring and maintenance procedures.	Technical proposal; operations procedures.	Delivered during implementation
41.	**Mobile Responsiveness**	Responsive interface for desktop, laptop, tablet and phone. BIRDS is also a progressive web app (PWA): it is installed on a phone, tablet or computer straight from the browser (Chrome/Edge "Install app"; Safari "Add to Home Screen" on iPhone/iPad), with no app store, and then opens full screen like a native app. There is no separate mobile app to maintain; installed copies update themselves when the server is updated.	Open the demo on a phone; user menu › "Install app".	Demonstrated
42.	**Browser Support**	Tested on current Google Chrome, Microsoft Edge and Mozilla Firefox.	Open the demo in any of the three browsers.	Demonstrated
43.	**Localization**	Complete English and Bangla interface with a language switch; user-entered content is stored and shown in either language.	EN / Bangla language switch in the header.	Demonstrated
44.	**Data Retention**	Configurable retention period and archiving age; records past retention are flagged for review and never deleted automatically.	Admin › Security & system › Archiving & retention.	Demonstrated
45.	**Data Archiving**	Administrators archive and restore records (manually or by a daily job); archived records keep their history and audit trail and remain searchable.	Incident page › Archive; Incidents › archived filter.	Demonstrated
46.	**Administrative Module**	Central Admin module for users, roles and permissions, organisational units, categories, workflow settings, security and system parameters, retention, e-mail and operations.	Administration (hq_admin); Security & system opens with the system status.	Demonstrated

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47.	**Configuration**	Categories, types, organisational hierarchy, permissions, password/session policy, overdue time, retention and report parameters are configurable without code changes. Workflow stages and notification rules become configurable during implementation.	Administration tabs.	Demonstrated; completed at deployment
48.	**Logging & Monitoring**	Application, HTTP and security logs are kept separately; the audit trail is viewable in the application. Log retention settings and the administrative monitoring view are completed at deployment.	Audit trail; Admin › Operations (marked DEMO).	Demonstrated; completed at deployment
49.	**Security Hardening**	Built following secure development practice against the OWASP Top 10: validated input, parameterised database access, access control on every request, secure headers, rate limits, no secrets in the browser, dependency audit. An OWASP Top 10 assessment is provided.	Security assessment (to be delivered).	Demonstrated; completed at deployment
50.	**Vulnerability Assessment**	Application security testing and vulnerability assessment before production deployment, with the report provided; critical and high-risk findings are fixed before acceptance.	Vulnerability assessment report (to be delivered).	Delivered during implementation
51.	**Source Code / Ownership**	All application components developed for BGB are delivered with full source code, database schema and migrations, configuration and technical documentation.	Source code handed over on contract.	Delivered during implementation
52.	**Documentation**	System Administration Manual, User Manual, Installation/Deployment Guide, API Documentation (already available in OpenAPI form), Database Documentation and Backup/Recovery Procedure.	/api/docs (API documentation).	Delivered during implementation
53.	**Training**	Hands-on training for system administrators, designated operators and end users covering incident entry, verification, reporting and administration, with training material.	Training plan (to be delivered).	Delivered during implementation

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54.	**Warranty & Support**	One year warranty and technical support: bug fixing, security patches, configuration support and troubleshooting.	Warranty terms in the commercial offer.	Delivered during implementation
55.	**Customization**	Categories, hierarchy, report templates and workflow settings are adjusted through the Admin module; further customisation of forms, reports and workflows to BGB's operational requirements is included in implementation.	Administration; Reports templates.	Demonstrated; completed at deployment
56.	**Deployment**	Installation, configuration, integration, testing, data migration where applicable, commissioning and production deployment on BGB infrastructure are included.	Installation/Deployment Guide (to be delivered).	Delivered during implementation
57.	**Acceptance Testing**	Functional acceptance test cases are prepared against the approved requirements and run with BGB; final acceptance follows their successful completion.	UAT test cases (to be delivered).	Delivered during implementation
58.	**Future Expansion**	Modular design with a documented REST API for other systems; already extended with secure messaging, voice calls and push notifications, and delivered as a progressive web app installable on phones from the browser, showing readiness for mobile, GIS and evidence-system integration.	Messages, Call log; /api/docs.	Demonstrated

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